INTERNSHIP REPORT

Submitted

For the Partial Fulfilment of the Requirement for the Degree of Bachelor of _____ with _____ (Major and Minor Subjects) under **Magadh University**

Submitted to

The Department of _____, _____ **(College Name)**

[Paste College Logo here]

Submitted by

Name : _____

Semester : _____

Programme FYUGP : _____

University's Registration No. : _____

College Roll No. : _____

Year : _____

RK NEXORA
CAMPUS TO CAREER
A Unit of RKNexora Private Limited.

INTERNSHIP REPORT

Topic - Supply Chain Management

Submitted to the Head of Department, _____ (College Name), in partial fulfilment of the requirements for the Degree of _____, Magadh University

Name of Student : _____

Programme : _____

University Registration No. : _____

College Roll No. : _____

Internship Registration No. : _____

Name of the IPO / Agency : RKNEXORA PRIVATE LIMITED

Internship Period : _____ to _____

Semester : _____

Year / Session : _____

RKNEXORA PRIVATE LIMITED
RKNEXORA
PRIVATE LIMITED
GAYA, BIHAR - 805236, INDIA

_____ _____

Signature of Student **Seal & Signature of IPO**

ACKNOWLEDGEMENT

I express my sincere gratitude to RK NEXORA, RKNEXORA PRIVATE LIMITED, _____ (College Name), my faculty mentors and the internship supervisor, _____, for their guidance and support. The structured activities, learning resources and regular feedback helped me complete this internship in Supply Chain Management successfully.

I am thankful to the Head of the Department and the college mentor for their encouragement and for guiding me in preparing this report. I am also grateful to the management of RKNEXORA PRIVATE LIMITED for providing a professional learning environment and the opportunity to learn and apply my knowledge in a practical setting. The concept sessions, hands-on tasks with sample data and case studies, and the supervisor's patient and detailed feedback helped me understand how goods and information move from suppliers to customers.

Finally, I thank my family and peers for their constant encouragement throughout the programme.

_____

Name: _____

University Registration No. - _____

INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by RKNEXORA PRIVATE LIMITED for the internship in Supply Chain Management is to be attached on this page.

[Attach the original / printed copy of the QR-verified Internship Completion Certificate here]

Certificate No. _____

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PREFACE

Internships are an important part of the undergraduate programme, as they allow students to see how classroom learning is applied in real organisations. The Four-Year Undergraduate Programme (FYUGP), introduced in line with the National Education Policy 2020, gives special importance to experiential learning, skill development and employability. This report has been prepared as part of the internship requirement of Magadh University and the Higher Education guidelines applicable in Bihar.

The report describes the internship undertaken in Supply Chain Management at RKNEXORA PRIVATE LIMITED, the plan followed, the work carried out, and the learning gained. Everything a person buys - a packet of biscuits, a mobile phone, a bag of cement - has travelled a long way before it reaches the shop. Raw materials have been bought, made into products, stored in warehouses, loaded on trucks and trains, and finally placed on a shelf. Supply chain management (SCM) is the planning and control of this whole journey so that the right goods reach the right place at the right time and at the right cost. Bihar, with its growing trade, agriculture and food-processing activity and its position between eastern and northern India, is an important link in many supply chains, and logistics is one of the fastest-growing fields of employment. This report describes how the main parts of supply chain management were studied and how an inventory and logistics study was prepared for a fictitious trading firm using dummy data.

The report has been arranged in the order suggested by the University, beginning with the cover page and acknowledgement and ending with the conclusion and attendance sheet. It is hoped that it gives a clear and honest account of the internship experience. All examples in this report use dummy data and fictitious names; no real personal or business information has been handled.

EXECUTIVE SUMMARY

This report documents the internship completed in Supply Chain Management at RKNEXORA PRIVATE LIMITED (RK NEXORA - 'Campus to Career'). The programme required a minimum of 120 hours, completed over about four to five weeks (28 working days) with daily sessions of four to five hours, and was supported by an attendance log, supervisor guidance and review.

Background. A supply chain is the network of suppliers, manufacturers, warehouses, transporters, retailers and customers that together produce and deliver a product. Supply chain management coordinates the flows within this network: the flow of goods forward to the customer, the flow of money, and the flow of information in both directions. A good supply chain keeps the cost low, the service level high and the stock at a sensible level. A poor one leads to stock-outs and lost sales on one side and unsold, damaged or expired stock on the other.

Objectives. The internship aimed to (i) understand the stages of the supply chain from sourcing to delivery, (ii) learn demand forecasting from past sales data, (iii) learn procurement and sourcing, (iv) learn inventory control, including safety stock, EOQ and ABC analysis, (v) learn warehousing operations, (vi) learn transportation and logistics, (vii) learn vendor evaluation, (viii) learn order fulfilment, (ix) understand supply chain software, and (x) identify ways to optimise cost.

Work carried out. Each topic had a concept session followed by hands-on calculations and exercises in MS Excel using dummy data and a fictitious trading firm. The student forecast demand with moving averages and exponential smoothing, calculated reorder levels, safety stock and the economic order quantity, classified items by ABC analysis, scored vendors, mapped the warehouse and order-fulfilment processes and compared transport options. The later part of the programme was devoted to the project, a case study exercise, a mock practical exercise, supervisor review and preparation of this report.

Project. The work was tied together in a project titled **Supply Chain & Inventory Optimization Study**. The study took a fictitious building-materials trader, Bihar Build Supplies, with ten main items, and examined its demand pattern, inventory classification, order quantity, safety stock and delivery cost, and then made recommendations to reduce stock-outs and inventory cost.

Method. A blended method was followed - learn, practise, document and review - supported by a daily attendance log with day-wise remarks and regular feedback from the supervisor.

Learning gained. The student gained a clear understanding of the supply chain and its stages, practical skill in forecasting, EOQ, safety stock and ABC analysis, vendor scoring and order-fulfilment steps, and the ability to read a supply chain problem in terms of cost, service and risk.

Key observations. Every part of a supply chain depends on the others; a forecast that is wrong by a small amount creates a large effect on stock; a small number of items (class A) usually represent most of the value; and savings in the supply chain come more from good planning and information than from cutting prices.

Conclusion. The internship met its objectives and contributed to the student's academic understanding, employability and confidence. The report follows the University's suggestive format: cover page, acknowledgement, certificate, contents, preface, executive summary, about the agency, plan and execution, learning outcome, conclusion and attendance sheet.

ABOUT THE AGENCY

RK NEXORA, a unit of RKNEXORA PRIVATE LIMITED, provides structured internship and skill-development programmes under the theme 'Campus to Career'. The organization focuses on academic-industry alignment, practical exposure, assessment, documentation and employability-oriented learning.

Particulars	Details
Name of Agency	RKNEXORA PRIVATE LIMITED
Programme / Unit	RK NEXORA - Campus to Career
Registered Office	C/O Mukesh Kumar, Khiri, PS Atri, Jethiyan, Atri, Gaya - 805236, Bihar, India
CIN	U85499BR2026PTC086757
Email	rknexorapvtltd@gmail.com
Website	www.rknexora.org
Internship Topic	Supply Chain Management

1. Vision

To support students in developing practical, professional and industry-relevant capabilities.

2. Mission

To deliver structured learning experiences that connect academic knowledge with workplace expectations.

3. Activities of the Agency

1. Structured internship programmes aligned with university and higher-education requirements.
2. Skill-development learning modules, practical activities and assignments.
3. Attendance tracking, supervision, assessment and documentation.
4. Employability-oriented guidance connecting campus learning to career readiness.

4. Programme Structure

The internship programme is organised around a fixed number of learning hours, completed over a defined period. Each intern is registered on the internship portal with a unique registration number, is assigned a supervisor, follows a day-wise plan and maintains an attendance log. Learning is organised into three phases: **Orientation** (Days 1-2), **Learning and Practice** (Days 3-22, ten sub-topics of two days each: one concept day and one hands-on day) and **Project and Review** (Days 23-28), followed by a concluding review.

5. Role of the Supervisor

The internship supervisor assigned by the agency guided the student through the programme, explained concepts, assigned practice tasks, reviewed the daily log and gave feedback. The supervisor also verified attendance and reviewed the student's work at the end of the programme.

PLAN AND EXECUTION

1. Objectives

The internship was planned around the following objectives, each linked to a part of the programme:

1. Understand the supply chain, its stages and the three flows of goods, money and information (Week 1).
2. Learn demand forecasting using past sales data (Week 1-2).
3. Learn procurement and sourcing, and vendor evaluation (Week 2).
4. Learn inventory control: stock levels, safety stock, reorder point, EOQ and ABC analysis (Week 2-3).
5. Learn warehousing operations, transportation and logistics, and order fulfilment (Week 3).
6. Understand supply chain software and cost optimisation (Week 4).
7. Complete an inventory optimisation study as a project and present the findings (Weeks 4-5).

2. Methodology

The programme followed a blended method made up of four connected steps, repeated for each topic:

Learn. Each topic began with a concept session in which the supervisor explained the idea, its main terms and how it is used in real organisations. Notes were prepared in the student's own words.

Practise. The concepts were then applied in guided hands-on tasks - forecasting monthly demand with moving averages, calculating reorder level and safety stock, working out the economic order quantity, classifying ten items by ABC analysis, scoring vendors on quality, price, delivery and service, drawing a warehouse flow, comparing transport modes and preparing a cost-saving list, all in MS Excel. Only sample data and fictitious examples were used, so that no real personal or business information was handled.

Document. Every day's work was recorded in the internship attendance log with a short remark, and sheets, files and screenshots were saved so that the work could be reviewed.

Review. The supervisor reviewed the work, pointed out errors and suggested improvements. Review days, a case study and a mock practical exercise helped to consolidate the learning.

3. Understanding Supply Chain Management - Concepts Studied

3.1 The supply chain and its stages

A supply chain begins with the suppliers of raw materials and ends with the customer who uses the product. In between are manufacturers or processors, who change the material into a product; warehouses, which store it; distributors and wholesalers, who break large lots into smaller ones; transporters, who carry the goods; and retailers, who sell to the end customer. Three kinds of flow run through the chain: the product flows forward, money flows backward as payment, and information - orders, forecasts and stock positions - flows in both directions. Supply chain management also includes the reverse flow of returns and repairs. The figure below shows the stages.

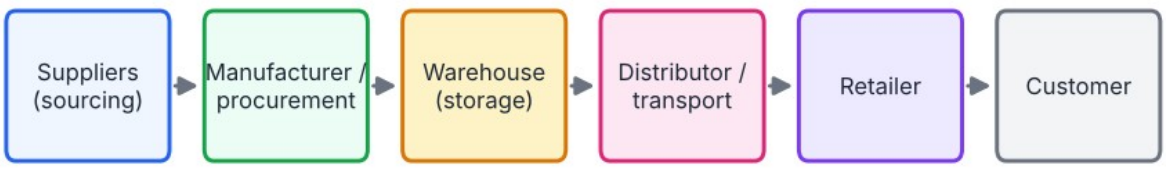


Figure 1: Stages of a supply chain

3.2 Objectives and performance of a supply chain

The two aims of a supply chain pull in different directions. One is to give the customer high service - the product is available, delivered on time and in good condition. The other is to keep the cost low - the cost of buying, storing, moving and handling the goods. Spending more on stock and fast transport raises service but also cost, and the manager's task is to find the right balance. The common performance measures are the fill rate (the percentage of order quantity supplied from stock), the on-time delivery percentage, the order cycle time, the inventory turnover (cost of goods sold divided by average inventory), the logistics cost as a percentage of sales and the perfect order rate (orders delivered complete, on time, undamaged and with correct documents).

3.3 Demand forecasting

All planning in the supply chain starts from a forecast of demand, because goods must be ordered and made before they are sold. Forecasting methods are of two kinds. Qualitative methods depend on judgement - the views of sales staff, customers and experts - and are used when there is no past data. Quantitative methods use past data. The simple methods studied were the moving average, which averages the last few periods, and exponential smoothing, which gives more weight to recent periods. The forecast should always be compared with what actually happened, and the error measured, so that the method can be improved.

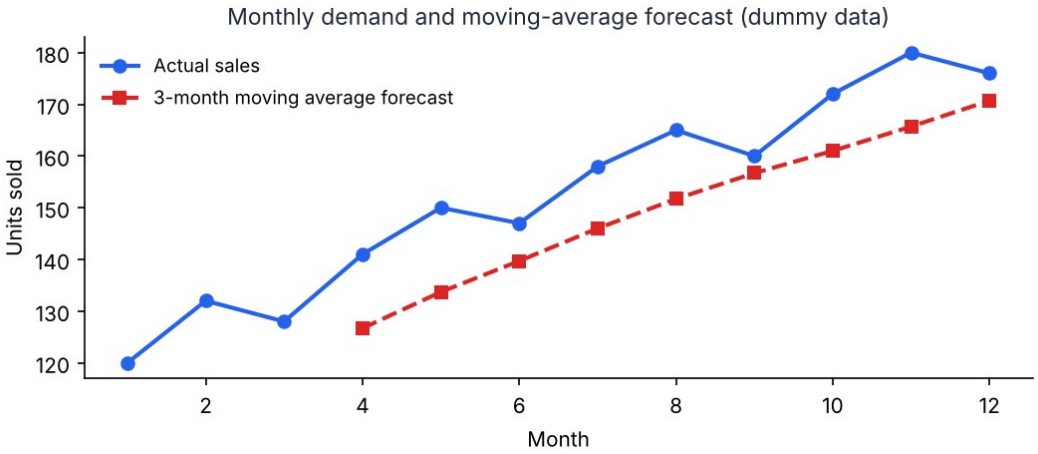


Figure 2: Actual demand and 3-month moving average forecast

For twelve months of dummy sales (120, 132, 128, 141, 150, 147, 158, 165, 160, 172, 180 and 176 units) the three-month moving average forecast for month 13 is (172 + 180 + 176) / 3 = 176 units. Applying the method to months 4 to 12 gave a mean absolute deviation (MAD) of 10.8 units and a mean absolute percentage error of 6.8%. Exponential smoothing with a smoothing constant of 0.3 gave a forecast of about 169 units for the same month. Because the demand has a rising trend, both methods lag behind the actual demand, which suggested that a method with a trend adjustment would be better. The exercise showed why a forecast is an estimate and not a promise, and why safety stock is needed.

3.4 Procurement and sourcing

Procurement is the process of obtaining the goods and services the business needs. Sourcing is the strategic part of it: deciding what to buy, from whom, at what price and on what terms. The usual steps are identifying the need, preparing a specification, finding suppliers, asking for quotations (request for quotation, RFQ), comparing and negotiating, issuing a purchase order, receiving and inspecting the goods, and paying the invoice. Important decisions are single or multiple sourcing, local or distant suppliers, and make or buy. For important items, buying from two or three suppliers reduces the risk of a stoppage, while for small items one reliable supplier may reduce the effort. In negotiation, the total cost - price, transport, quality, credit period and wastage - should be compared, and not the price alone.

3.5 Inventory control: reorder level, safety stock and EOQ

Inventory is the stock of goods held to meet demand. It is needed because demand and supply are uncertain, but it costs money: capital is locked up, space is used, and goods may be damaged, stolen or become obsolete. The aim is to hold enough but not too much. The reorder point is the stock level at which a new order is placed; it equals the demand during the lead time plus a safety stock. The safety stock protects against a higher demand or a late delivery and is often calculated as z x (standard deviation of daily demand) x square root of lead time, where z depends on the service level wanted (1.65 for 95%).

The economic order quantity (EOQ) is the order size that makes the total of ordering cost and holding cost the lowest. The formula is EOQ = square root of (2 x D x S / H), where D is the yearly demand, S is the cost of placing one order and H is the holding cost per unit per year. For cement of a dummy firm, D = 12,000 bags, S = Rs. 500 and H = Rs. 12, so EOQ = square root of (2 x 12,000 x 500 / 12) = 1,000 bags. The firm should therefore place 12 orders in a year. The yearly cost is (12 x Rs. 500) + (500 x Rs. 12) = Rs. 6,000 + Rs. 6,000 = Rs. 12,000, and at the EOQ the ordering and holding costs are equal. For safety stock, with a daily demand deviation of 4 bags and a lead time of 9 days, safety stock = 1.65 x 4 x 3 = about 20 bags.

3.6 ABC analysis

Not all items deserve the same attention. ABC analysis ranks items by their yearly consumption value (quantity x price) and divides them into three classes: A items, which are few but account for about 80% of the value, B items, the next 15%, and C items, which are many but account for only about 5%. Class A items get tight control, frequent review and careful forecasting; class B items get normal control; and class C items are bought in bulk with simple rules. The analysis was carried out for ten items of the dummy firm, with a total yearly consumption value of Rs. 13,00,000.

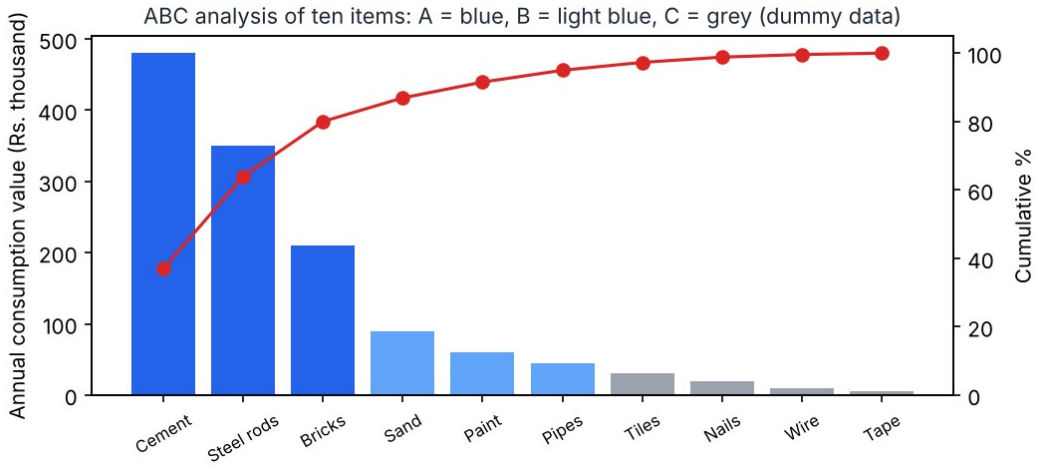


Figure 3: ABC analysis of ten items

Cement (36.9%), steel rods (26.9%) and bricks (16.2%) together account for 80% of the value and are class A, although they are only 3 of the 10 items. Sand, paint and pipes bring the total to 95% and are class B, and the remaining four items, tiles, nails, wire and tape, are class C with only 5% of the value. The firm should therefore spend its management time on the first three items and manage the last four with simple rules.

3.7 Warehousing operations

A warehouse is not just a store; it is a place where goods are received, checked, stored, picked, packed and dispatched. The main activities are receiving (unloading, counting and inspecting goods against the purchase order), put-away (placing goods in the right location), storage (arranging goods by item code, size, weight and the first-expired-first-out rule), order picking (collecting the goods for an order), packing and dispatch. Good warehouse practice includes a clear layout with separate zones, location codes for every rack, regular cycle counting instead of one yearly count, safety rules for lifting and fire, and records that agree with the physical

stock. A warehouse management system (WMS) uses barcodes and software to guide these tasks and reduce errors.

3.8 Transportation and logistics

Transportation moves goods between the points of the chain, and logistics includes transport, warehousing, packing and handling. The modes are road, rail, air and water (and pipelines for fluids). Road transport is flexible and reaches every place, and is the main mode for short and medium distances; rail is cheap for heavy and bulky goods over long distances; water is the cheapest for very bulky goods; and air is the fastest but the most expensive, and is used for urgent and valuable goods. The choice depends on cost, speed, reliability, the nature of goods and the places to be served. Freight is charged by weight, volume and distance, and delivery tracking by GPS and mobile apps gives the customer the position of the goods.

Mode	Speed	Cost per tonne-km	Best suited for	Limitation
Road	Medium	Medium	Door-to-door, short and medium distance	Traffic, fuel cost, limited load
Rail	Slow to medium	Low	Heavy, bulky goods over long distances	Fixed routes and timetables
Water	Slow	Very low	Very bulky goods such as coal and grain	Limited routes, slow
Air	Very fast	High	Urgent, valuable, light goods	Cost, weight limits

3.9 Vendor evaluation

The performance of suppliers directly affects the performance of the buyer, so suppliers should be rated regularly and not only when there is a problem. A vendor rating uses criteria such as quality (percentage of goods accepted), price (compared with the market), delivery (percentage of orders on time), service (response to problems) and documentation. Each criterion is given a weight and a score, and the weighted score ranks the vendors. In the practice exercise, three vendors of steel rods were scored out of 10 on quality (weight 40%), price (30%), delivery (20%) and service (10%). Vendor A scored (9 x 0.4) + (7 x 0.3) + (8 x 0.2) + (8 x 0.1) = 8.1, Vendor B scored (7 x 0.4) + (9 x 0.3) + (6 x 0.2) + (7 x 0.1) = 7.4 and Vendor C scored (8 x 0.4) + (8 x 0.3) + (9 x 0.2) + (6 x 0.1) = 8.0. Vendor A was ranked first, with Vendor C close behind, so Vendor C could be kept as a second source.

3.10 Order fulfilment, supply chain software and cost optimisation

Order fulfilment is the whole process from receiving a customer's order to the confirmation of delivery. The steps are order entry and checking, stock availability check, picking, packing, invoicing, dispatch, delivery and confirmation, and the handling of returns. The key measures are the order cycle time, the fill rate and the perfect order rate.

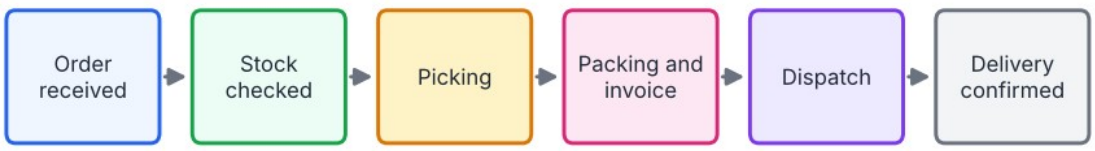


Figure 4: Steps of order fulfilment

Supply chain software helps to carry out these processes. A supply chain management (SCM) system plans demand, supply and production; a warehouse management system (WMS) controls stock and tasks inside the

warehouse; and a transportation management system (TMS) plans routes, compares freight and tracks vehicles. They are often connected with the ERP system. The cost of the supply chain can be reduced by better forecasting, ABC-based control, ordering at the EOQ, consolidating loads, choosing the right transport mode, reducing damage and returns, negotiating with suppliers on total cost and removing steps that add no value.

3.11 Supply chain risks and resilience

A supply chain can fail at any link: a supplier may close, a truck may be delayed by floods, a price may suddenly rise or a customer may cancel an order. Bihar's supply chains face special risks such as monsoon floods that cut roads and rail links, and the concentration of many goods through a few trading hubs. Supply chain risk management first identifies the risks, then estimates their likelihood and effect, and then plans actions to reduce them. The student studied the main kinds of risk and the usual responses.

Risk	Example	Possible response
Supply risk	The only supplier of steel rods stops supply	Second source; safety stock of key items
Demand risk	Sales fall suddenly after a price rise	Review the forecast monthly; flexible orders
Transport risk	Road closed by floods in the monsoon	Alternative routes and early stocking before the season
Quality risk	A batch of cement is found damp	Inspection at receiving; supplier quality agreements
Financial risk	A customer delays payment for three months	Credit limits and regular follow-up
Information risk	Records are lost or are not updated	Backups, shared records and regular counts

A resilient supply chain keeps some extra capacity or stock for the important items, has more than one source and route, shares information with suppliers and customers and reviews the plan regularly. The student noted that resilience has a cost, and that the aim is not to remove every risk but to be ready for the most likely and most damaging ones.

4. Week-wise Plan and Execution

Week	Days	Activities
Week 1	Days 1-6	Orientation and foundations: company induction, supply chain overview, demand forecasting and procurement and sourcing, each with a study session and a practice task.
Week 2	Days 7-12	Inventory and vendors: inventory control with EOQ, safety stock and ABC analysis, warehousing operations and vendor evaluation.
Week 3	Days 13-18	Movement and fulfilment: transportation and logistics, order fulfilment and supply chain software.
Week 4	Days 19-24	Optimisation: cost optimisation, the Supply Chain & Inventory Optimization project, a case study exercise and a mock practical exercise.
Week 5	Days 25-28	Project closure: supervisor review and feedback, internship report drafting, presentation preparation and final review.

5. Week-wise Narrative

Week 1: The first week was devoted to orientation and foundations. After company induction, the student studied the stages and flows of the supply chain, practised forecasting with moving averages and exponential smoothing, and studied procurement and sourcing, including RFQs and purchase orders.

Week 2: In the second week the student studied inventory control and calculated reorder points, safety stock and the EOQ, and classified items by ABC analysis. The week then moved to warehousing operations and to vendor evaluation, with a weighted rating of three vendors.

Week 3: The third week was devoted to transportation and logistics, in which the transport modes were compared, and to order fulfilment and supply chain software, including WMS and TMS.

Week 4: In the fourth week the student studied cost optimisation and completed the inventory optimisation study, a case study exercise and a mock practical exercise.

Week 5: The last week was devoted to project closure: supervisor review and feedback, drafting of the internship report, preparation of a presentation and final review before submission.

6. Topic-wise Learning in Detail

6.1 Supply Chain Overview

Concept: The stages of the supply chain from sourcing to delivery. **Practice carried out:** Drawing the supply chain of a packet of rice from the farmer to the customer and marking the three flows.

6.2 Demand Forecasting

Concept: Forecasting methods using past sales data. **Practice carried out:** Calculating three-month moving average and exponential smoothing forecasts for twelve months of data and measuring the error (section 3.3).

6.3 Procurement and Sourcing

Concept: Supplier selection, RFQs, purchase orders and negotiation. **Practice carried out:** Comparing three quotations on total cost, preparing an RFQ and a purchase order and writing a negotiation plan.

6.4 Inventory Control

Concept: Stock levels, safety stock, ABC analysis and stock records. **Practice carried out:** Calculating EOQ, reorder point and safety stock, classifying ten items by ABC analysis and preparing a stock ledger.

6.5 Warehousing Operations

Concept: Receiving, put-away, picking, packing and dispatch. **Practice carried out:** Drawing a warehouse layout with zones, writing receiving and picking checklists and a rack location code scheme.

6.6 Transportation and Logistics

Concept: Transport modes, freight, routes and delivery tracking. **Practice carried out:** Comparing road, rail and air for three consignments on cost and time, and planning a delivery route for six customers.

6.7 Vendor Evaluation

Concept: Vendor rating on quality, price, delivery and service. **Practice carried out:** Scoring three vendors on four weighted criteria and ranking them (section 3.9).

6.8 Order Fulfilment

Concept: Order processing, picking, shipment and delivery confirmation. **Practice carried out:** Mapping the order-to-delivery process, measuring the order cycle time and listing causes of delay.

6.9 Supply Chain Software

Concept: SCM, WMS and TMS tools and their features. **Practice carried out:** Comparing the features of SCM, WMS and TMS tools and preparing a note on which a small trader needs first.

6.10 Cost Optimization

Concept: Ways to reduce logistics, inventory and procurement costs. **Practice carried out:** Preparing a list of ten cost-saving ideas with the likely saving and effort for each.

7. Good Practice, Safety and Ethics - Points Studied

A supply chain connects many organisations and many people, so honesty, safety and care for the environment are part of its management. The points studied are summarised below.

- **Ethical sourcing:** buy from suppliers who follow the law, pay fair wages and do not use child or forced labour.
- **Fair dealing:** pay suppliers on time and follow the agreed terms; avoid bribes and unfair pressure.
- **Quality and safety:** check goods at receiving, store them safely and follow the safety rules for lifting, fire and hazardous goods.

- **Accurate records:** stock, delivery and invoice records must match the physical goods; changes must be authorised.
- **Legal compliance:** GST and e-way bill rules, the Motor Vehicles Act rules on loading, and food and drug licence rules must be followed.
- **Environmental care:** reduce waste and packaging, plan routes to save fuel and dispose of damaged goods properly.
- **Confidentiality:** prices, forecasts and supplier information are business secrets and must not be shared.

8. Process Flows Studied

8.1 Procure-to-pay in the supply chain

- Need identified and purchase requisition approved.
- Suppliers asked for quotations; the best total-cost offer selected.
- Purchase order issued with quantity, price, delivery date and terms.
- Goods received and checked against the order; a goods receipt note prepared.
- Invoice matched with the order and receipt, and payment made on the due date.

8.2 Receiving goods in a warehouse

- Check the vehicle, the delivery challan and the purchase order.
- Unload and count the goods; inspect them for damage and quality.
- Record the receipt and update the stock; reject or return wrong goods.
- Put the goods away in the correct location and record the location code.
- Report differences to purchase and to the supplier the same day.

9. Worked Examples with Sample (Dummy) Data

Worked example 1 - reorder level and EOQ for three items. For the dummy firm, the average daily demand, the lead time, the safety stock and the order cost were used to find the reorder points and order quantities. Reorder point = (daily demand x lead time) + safety stock.

Item	Yearly demand	Daily demand	Lead time (days)	Safety stock	Reorder point	EOQ
Cement bags	12,000	40	9	20	380	1,000
Steel rods (kg)	30,000	100	7	80	780	2,500
Bricks (nos.)	1,20,000	400	5	300	2,300	8,000

For cement, the reorder point is (40 x 9) + 20 = 380 bags: when the stock falls to 380 bags, an order of 1,000 bags is placed, and it arrives in about nine days, by which time the stock should have fallen to about 20 bags, the safety stock. The firm had earlier ordered cement in lots of 3,000 bags, twice a year; at the EOQ of 1,000 bags the yearly ordering and holding cost fell from Rs. 20,000 to Rs. 12,000. This is a saving of Rs. 8,000 on one item.

Worked example 2 - cost of ordering in lots of 3,000. With D = 12,000, a lot of 3,000 gives 4 orders a year, so the ordering cost is 4 x Rs. 500 = Rs. 2,000, and the average stock is 1,500 bags, so the holding cost is 1,500 x Rs. 12 = Rs. 18,000. The total is Rs. 20,000, which is Rs. 8,000 more than the Rs. 12,000 at the EOQ. The exercise showed that the total cost curve is flat near the EOQ, so a small rounding of the order size does not make much difference, but a very large lot is costly.

Worked example 3 - order fulfilment time. For a dummy customer order, the time at each step was noted: order entry 0.5 hour, stock check 0.5 hour, picking 2 hours, packing 1 hour, invoicing 0.5 hour, loading and dispatch 1.5 hours and transport 8 hours. The order cycle time is 14 hours. Picking and transport were the largest steps; the study suggested a better rack arrangement for faster picking and fixed dispatch times in the morning to avoid evening delay.

Worked example 4 - service level and safety stock. Safety stock depends on the service level wanted. For the cement example (daily demand deviation 4 bags, lead time 9 days), the safety stock for different service levels is calculated below using the safety stock = z x 4 x 3 formula.

Service level	z value	Safety stock (bags)	Chance of stock-out in a cycle
90%	1.28	15	10%
95%	1.65	20	5%
99%	2.33	28	1%

Raising the service level from 95% to 99% needs only about 8 more bags for cement, which is cheap, but for an expensive item with high variation the same step could lock up a large amount of money. The student therefore learnt that the service level should be decided item by item - high for class A items that customers cannot do without, and lower for class C items - and not as a single figure for all items.

10. Project: Supply Chain & Inventory Optimization Study

Problem statement. Bihar Build Supplies, a fictitious building-materials trader in Patna with ten main items, a warehouse and five delivery vehicles, faces two problems: the shop sometimes runs out of cement and steel, and at other times it holds large quantities of slow-moving items such as tiles and paint. Deliveries to sites are often late. The aim of the project was to study the supply chain and suggest improvements.

10.1 Method

- Collect twelve months of dummy sales data and the yearly consumption values of ten items.
- Forecast demand by moving average and exponential smoothing and measure the error.
- Classify the items by ABC analysis.
- Calculate the EOQ, safety stock and reorder point for class A items.
- Rate three steel suppliers on quality, price, delivery and service.
- Map the order fulfilment process and find the causes of delay.
- Estimate the savings from the recommended changes.

10.2 Findings

Area	Finding	Evidence
Demand	Rising trend of about 5 units per month; forecast error 6.8%	Moving average of 12 months
Inventory	Three items (30%) are class A with 80% of value; four class C items take 4 stock locations but only 5% of value	ABC analysis
Stock-outs	Cement and steel were reordered by 'feel' and not by reorder point	Stock records
Order size	Cement lots of 3,000 bags are three times the EOQ	EOQ calculation
Vendors	Vendor A is best overall; Vendor C is a good second source	Vendor rating
Delivery	Order cycle time of 14 hours, with picking and transport as the largest steps	Process timing

10.3 Recommendations and expected savings

- Use a reorder point and the EOQ for class A items, with a weekly review of their stock.
- Place cement orders of 1,000 bags (EOQ) instead of 3,000, saving about Rs. 8,000 a year on this item alone in ordering and holding cost.
- Keep a safety stock of 20 bags of cement and 80 kg of steel rods, so that the chance of a stock-out falls to about 5%.
- Buy class C items in bulk once or twice a year with a simple two-bin system.
- Develop Vendor C as a second source of steel rods to reduce the risk of a supply failure.
- Re-arrange the warehouse so that class A items are near the dispatch area, and fix morning dispatch times for delivery routes.

10.4 Limitations

The study used dummy data and simple methods. A real study would need more data, a check for seasonality (building work slows in the monsoon), the cost of storage space and the effect of price changes. The student noted that the numbers are for learning, and that real decisions should be taken only after checking the real data and discussing with the people who run the business.

10.5 Implementation plan and measures

Action	Time	Measure of success
Set EOQ and reorder points for class A items	Week 1	No stock-out of cement and steel in the month
Start weekly review of class A stock	Week 1	Review done every Monday
Re-arrange the warehouse	Weeks 2-3	Picking time reduced by 25%
Rate vendors every quarter	Quarterly	Vendor scores recorded; second source for steel

Action	Time	Measure of success
Fix morning dispatch and route plan	Week 2	On-time delivery above 90%
Review the forecast and fill rate monthly	Monthly	Fill rate above 95%; forecast error below 7%

11. Case Study Exercise: Stock-outs and Excess Stock in a Wholesale Distributor

A fictitious wholesale distributor of packaged foods in Muzaffarpur supplies 150 retail shops. The owner complains of two problems at once: popular items such as biscuits and soap are often out of stock, causing complaints and lost orders, while the warehouse is full of slow items, some of which have passed their best-before date. Orders are placed by the store keeper on 'feel', deliveries are made when the vehicle is full, and records are kept in a register. The student was asked to analyse the case.

Analysis. The problem is not too little or too much stock in total, but the wrong stock. There is no forecast, no classification of items, no reorder point and no tracking of expiry dates. Deliveries are planned for the convenience of the vehicle and not for the customer.

Proposed solution. Classify the items by ABC analysis and track class A items daily; forecast weekly demand from past sales; set reorder points and safety stock for class A and B items; use the first-expired-first-out rule and a weekly expiry report; plan delivery routes by area and fixed days; and record stock in a simple software or spreadsheet. A monthly review of the fill rate, stock turnover and expired stock would show whether the changes work.

Expected benefit and cautions. The fill rate of fast items would rise, the loss from expiry would fall, and the working capital locked in slow stock would be released. The cautions noted were that the changes should be introduced in steps, that the store staff must be trained, that supplier lead times must be checked and that the forecast should be reviewed every month.

12. Mock Practical Exercise: Preparing an Inventory Control Plan for a Small Trader

A mock practical exercise was conducted at the end of the learning phase to check that the student could prepare an inventory control plan on his own. The task was to prepare a plan for a fictitious stationery wholesaler with eight items. The steps followed were:

1. Data: the yearly sales quantity, price and lead time of the eight items were arranged in a table.
2. ABC analysis: the consumption value was calculated, the items were ranked and classified.
3. Forecast: a moving average forecast was prepared for the three class A items.
4. Order policy: EOQ, safety stock and reorder point were calculated for class A items.
5. Vendor choice: two vendors for the main item were rated on four criteria.
6. Warehouse: a simple layout was drawn with class A items near the dispatch door.
7. Cost and benefit: the expected saving in ordering and holding cost was estimated.
8. Presentation: the plan was explained to the supervisor in about five minutes.

13. Sample Checklists Prepared

Inventory control checklist

- Items classified by ABC analysis and reviewed every quarter.
- Reorder point and EOQ set for class A and B items.
- Stock register agrees with physical stock after each cycle count.
- Slow-moving, damaged and near-expiry items reported every month.

Warehouse and delivery checklist

- Goods checked against the purchase order at receiving; differences reported the same day.
- Every rack and bin has a location code; goods are stored by the first-expired-first-out rule.
- Safety rules for lifting, fire and stacking are followed.
- Delivery route, vehicle load and documents (invoice, e-way bill) are checked before dispatch.

14. Tools and Resources

The following tools and resources were used during the internship: MS Excel for forecasting, EOQ, ABC analysis, vendor scoring and charts, MS Word and MS PowerPoint for notes and presentation, templates for purchase orders, goods receipt notes and stock ledgers, case studies and learning material given by the agency, reference notes on GST, e-way bill and transport rules, descriptions of SCM, WMS and TMS tools, and regular mentor feedback.

15. Programme at a Glance

The programme was in the domain of Supply Chain Management with RKNEXORA PRIVATE LIMITED as the agency. It required a minimum of 120 hours over 28 learning days, organised in three phases: Orientation (Days 1-2), Learning and Practice (Days 3-22) and Project and Review (Days 23-28). Attendance was recorded in the Internship Attendance Log, signed by the supervisor (see Attendance Sheet).

The approximate distribution of the effort across the topics is shown below.

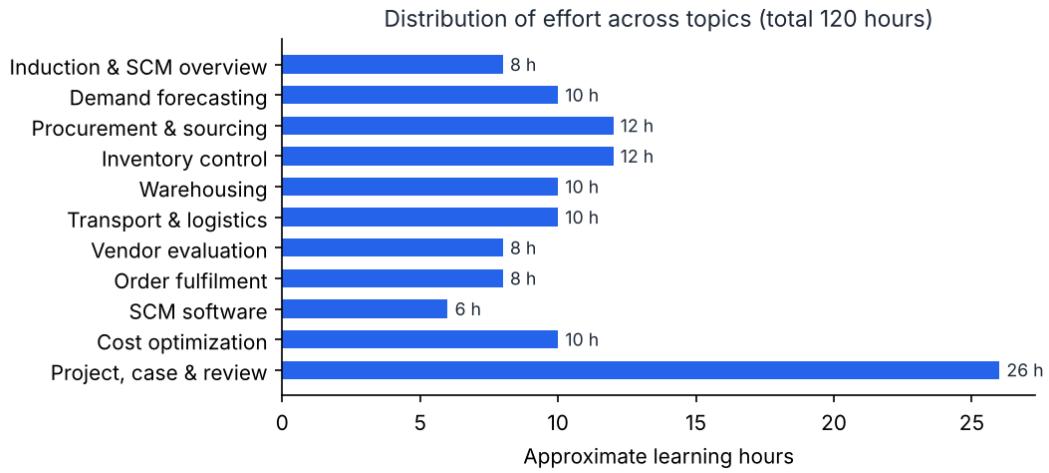


Figure 5: Distribution of learning effort across topics (120 hours)

LEARNING OUTCOME

a. Knowledge Acquired

1. The supply chain, its stages and the flows of goods, money and information.
2. The aims of supply chain management and measures such as fill rate, order cycle time and inventory turnover.
3. Forecasting with moving average and exponential smoothing, and the measurement of forecast error.
4. Procurement steps, sourcing decisions, RFQs and purchase orders.
5. Reorder point, safety stock, EOQ and ABC analysis.
6. Warehouse operations: receiving, put-away, storage, picking, packing and dispatch.
7. Transport modes, freight and route planning.
8. Vendor rating with weighted criteria.
9. Order fulfilment steps, and the functions of SCM, WMS and TMS software.
10. Ways of reducing logistics, inventory and procurement cost.

Glossary of Terms Learned

- **Supply chain** - The network that moves a product from suppliers to customers.
- **SCM** - Supply Chain Management.
- **Logistics** - Transport, storage and handling of goods.
- **Demand forecast** - An estimate of future sales.
- **Moving average** - Average of the last few periods used as a forecast.
- **Lead time** - Time between placing an order and receiving the goods.
- **Safety stock** - Extra stock kept against uncertainty.
- **Reorder point** - Stock level at which an order is placed.
- **EOQ** - The order size with the lowest total of ordering and holding cost.
- **ABC analysis** - Classifying items by value into A, B and C.
- **RFQ** - Request for Quotation.
- **Fill rate** - The share of demand supplied from stock.
- **Stock-out** - The situation when an item is not available.
- **FIFO / FEFO** - First in first out / first expired first out.
- **WMS / TMS** - Warehouse / Transportation Management System.
- **Cross-docking** - Moving goods directly from receiving to dispatch without storage.
- **Freight** - The charge for carrying goods.
- **Perfect order** - An order delivered complete, on time, undamaged and with correct documents.

b. Skills Learned

Technical skills. Forecasting demand with moving average and exponential smoothing; calculating reorder point, safety stock and EOQ; classifying items by ABC analysis; rating vendors with weighted scores; preparing purchase orders and goods receipt notes; planning warehouse layouts and delivery routes; comparing transport modes; and building tables and charts in MS Excel.

Professional skills. Analytical and logical thinking; accuracy with numbers; planning and coordination; negotiation and communication; attention to cost and detail; time management; and presenting recommendations in simple words.

These skills were developed through repeated practice and daily documentation. Working with dummy data from forecast to delivery built numerical accuracy, comparing options built decision skill, and explaining the study to the supervisor built clarity of communication.

c. Observed Attitude and Gained Values

1. **Honesty and fairness:** dealing with suppliers and customers on fair terms is the basis of a lasting supply chain.
2. **Responsibility for goods and records:** every item received, stored and delivered must be accounted for.
3. **Safety consciousness:** safe handling of goods and vehicles protects people and property.
4. **Cost awareness without cutting corners:** savings must never be made at the cost of quality or safety.
5. **Discipline and punctuality:** regular reporting at fixed timings and daily attendance records.

d. Challenges Faced

Understanding the many terms such as reorder point, safety stock and EOQ was handled by making a glossary and working each formula on one item.

Seeing why forecasts differed from actual demand was handled by plotting both on a chart and calculating the error.

Choosing the weights in vendor rating was difficult at first; it was handled by discussing the importance of each criterion with the supervisor and testing different weights.

Classifying items for ABC analysis without mistakes was handled by sorting and checking the cumulative percentage column.

Understanding warehouse activities was handled by drawing the flow of goods and a layout.

Managing time between calculations, documentation and report writing was handled by dividing work week by week and using review days for corrections.

e. Career Pathways

The knowledge gained supports careers such as Supply Chain Executive, Logistics Executive, Procurement and Purchase Executive, Inventory and Stores Executive, Warehouse Supervisor, Demand Planning Assistant, Vendor Management Executive and Operations Executive. Opportunities exist in manufacturing, retail, e-commerce, food and pharma distribution, logistics companies, construction and start-ups. Related higher-study options include MBA or PGDM in operations or supply chain management, BBA in logistics, and certificate courses in supply chain, logistics and warehouse management. A good next step is to learn Excel and one ERP or WMS tool, and to understand GST, e-way bill and transport documents.

CONCLUSION

The internship in Supply Chain Management was a valuable learning experience that integrated academic knowledge with practical activity. The programme achieved its objectives through structured learning, guided practice, assignments, supervisor review and documentation.

The student gained a clear understanding of how goods and information move from suppliers to customers - through forecasting, procurement, inventory control, warehousing, transport, vendor evaluation and order fulfilment - and acquired practical skill in calculating EOQ, safety stock and ABC classes and in preparing an inventory optimisation study. The experience has strengthened the student's employability and provides a foundation for further study and work in supply chain and logistics.

Reflection

Looking back, the most useful lesson of the internship was that a supply chain is a system in which small decisions have large effects. A forecast that is a little low leads to a stock-out; an order placed in too large a lot locks up money; a poorly arranged warehouse slows every order. Calculating the EOQ, the safety stock and the ABC classes with real-looking numbers made the ideas practical, and the case studies showed that good supply chain management is mostly careful planning, good records and good relations with suppliers.

Objectives and Achievement

The objective of understanding the supply chain was achieved through the stage diagram and the supply chain of rice. The objective of forecasting and procurement was achieved through the moving average exercise and the RFQ practice. The objective of inventory control was achieved through the EOQ, safety stock and ABC analysis. The objective of warehousing, transport and fulfilment was achieved through the layout, the mode comparison and the order cycle map. The objective of cost optimisation was achieved through the Bihar Build Supplies study and its recommendations.

Future Scope

The knowledge gained can support further study of operations and supply chain management and careers in logistics, procurement and inventory planning. With the growth of e-commerce, GST-based formal trade, new highways, rail freight corridors and warehousing in Bihar and eastern India, the demand for people who can plan and control the movement of goods is increasing.

Suggestions for Improvement of the Programme

1. Include a visit to a warehouse or a distribution centre with appropriate permissions.
2. Provide practice on a WMS or ERP inventory module with sample data.
3. Add a short session on GST, e-way bill and transport documents.
4. Continue regular mentor feedback and progress reviews.
5. Encourage students to maintain detailed daily documentation.

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7. Heizer, J. and Render, B. Operations Management. Pearson.
8. Government of India. The Central Goods and Services Tax Act, 2017 and the e-way bill rules.
9. Government of India. National Logistics Policy, 2022.
10. RK NEXORA internship programme handbook.

ATTENDANCE SHEET

The Internship Attendance Log issued by the IPO, with the supervisor's signature for each day, is to be attached on this page.

[Attach the signed Internship Attendance Log here]